is not completely specified by its location. Attached to the electron is an extra degree of freedom called its *spin*. Naively, the spin can be pictured as a little arrow that points in some direction, but that naive picture is too classical to accurately represent the real situation. The spin of an electron is about as quantum mechanical as a system can be, and any attempt to visualize it classically will badly miss the point.

We can and will abstract the idea of a spin, and forget that it is attached to an electron. The quantum spin is a system that can be studied in its own right. In fact, the quantum spin, isolated from the electron that carries it through space, is both the simplest and the most quantum of systems.

The isolated quantum spin is an example of the general class of simple systems we call qubits—quantum bits—that play the same role in the quantum world as logical bits play in defining the state of your computer. Many systems—maybe even all systems—can be built up by combining qubits. Thus in learning about them, we are learning about a great deal more.

1.3 AN EXPERIMENT

Let's make these ideas concrete, using the simplest example we can find. In the first lecture of *Volume I*, we began by discussing a very simple deterministic system: a coin that can show either heads (H) or tails (T). We can call this a two-state system, or a bit, with the two states being H and T. More formally we invent a "degree of freedom" called σ that can take on two values, namely +1 and −1. The state H is replaced by

$$\sigma = +1$$

and the state T by

$$\sigma = -1.$$

Classically, that's all there is to the space of states. The system is either in state $\sigma = +1$ or $\sigma = -1$ and there is nothing in between. In quantum